MILITARY SPECIFICATION

COMPUTER, AIR DATA CP-1005A/A

This specification is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The Computer, Air Data, CP-1105A/A covered by this specification will supply impact pressure, static pressure, barometric altitude, Mach number, and true airspeed information to cockpit indicators and flight control systems for aircraft control.

1.2 Associated equipment. This equipment shall operate with the associated equipment listed in 6.7.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

Federal

PPP-B-636	Boxes, Fiberboard
TT-L-20	Lacquer, Camouflage

Military

MIL-P-116	Preservation-Packaging, Methods of
MIL-T-152	Treatment, Moisture and Fungus Resistant, of Communications, Electronic and Associated Electrical Equipment, General Process for
MIL-C-172	Cases; Bases, and Mounts, Vibration (For Use with Electronic Equipment in Aircraft)
MIL-W-5088	Wiring Aerospace Vehicle
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-T-5422	Testing Environmental, Airborne Electronic and Associated Equipment
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of.
MIL-T-18303	Test Procedures; Preproduction, Acceptance and Life, For Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment Including Ground Support Equipment.
MIL-S-19500	Semiconductor Devices, General Specification for
MIL-S-20708/8	Synchro Control Transmitter, Type 26V-11CX4C

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STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-701	List of Standard Semiconductor Devices
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-2074	Failure Classification for Reliability Testing
MS33656	Fitting, End, Standard Dimensions for Flared Tube Connection and Gasket Seal

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CONRAC CORPORATION DOCUMENTS

7102432	Acceptance Test Procedure for Air Data Computer, P/N 560T58-1
7102530	Burn-In Procedure for Air Data Computer, P/N 560T58-1

DRAWINGS

560T58-1	Air Data Computer
7603729	Static Fitting
66442	Connector

(Application for copies of CONRAC Corporation documents should be addressed to CONRAC Corporation, Systems-East Division, 32 Fairfield Place, West Caldwell, New Jersey 07006.)

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AFCRC-TR-59-267 Air Research and Development command (ARDC) Model Atmosphere, 1959

(Application for copies of AFCRC-TR-59-267 should be addressed to the Air Force Geophysics Research Directorate, Cambridge Research Center, Bedford, Massachusetts.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish equipment samples for first article inspection and approval (see 4.4 and 6.3).

3.2 Parts and materials. In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

a. Parts and materials requirements shall conform to MIL-E-5400.

b. Nonrepairable subassemblies, as outlined in MIL-E-5400, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable.

c. When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard parts and material approval. Approval for the use of nonstandard parts and materials shall be obtained as outlined in MIL-E-5400.

3.2.2 Electron devices. Integrated circuits, transistors, and diodes shall be chosen and applied. Complements reports shall be prepared as outlined in MIL-STD-701 for procuring activity review and approval prior to preproduction testing. Diodes shall be selected in accordance with MIL-STD-701. Transistors shall be selected in accordance with MIL-S-19500.

3.3 Design and construction. The equipment shall conform with all the applicable requirements of MIL-E-5400 for design, construction, and workmanship, unless otherwise specified herein. The following air data tables shall be used in the functional design of the air data computer: AFCRC-TR-59-267, ARDC Model Atmosphere, 1959 Air Force Surveys in Geophysics No. 115.

3.3.1 Total weight. The total weight of the equipment excluding cables shall be a minimum consistent with good design. The weight shall not be greater than 25 pounds.

3.3.2 Reliability. The contractor shall conduct a reliability program in accordance with MIL-STD-785. On a reorder from a supplier who is producing the equipment, the program previously approved may be continued unless otherwise indicated in the contract or purchase order (see 6.2.1d).

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3.3.2.1 Operational stability. The equipment shall operate with satisfactory performance, continuously or intermittently for a period of not less than 2000 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating life. The equipment shall have a total operating life of 3000 hours with minimal servicing and replacement of parts. A list of parts requiring scheduled replacement due to wear during the life of the equipment, and the wearout life of such parts, shall be made available for the procuring activity review and approval (6.2.2).

3.3.2.3 Reliability in mean time between failure (MTBF). The equipment, including any Built-in-Test provisions, shall have 2000 hours of mean (operating) time between failures (θ_o as defined in MIL-STD-781) when tested and accepted as outlined under the requirements of 4.4.3.

3.3.2.4 Time totalizing meter. The equipment shall contain a 10,000 hour time totalizing meter in accordance with MIL-M-7793. The location of the time totalizing meter shall be on the front chassis plate.

3.3.3 Cabling and connectors.

3.3.3.1 Cables and connectors. The equipment shall provide for use of cables and connectors in accordance with MIL-E-5400.

3.3.3.2 Interconnecting cabling. The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plug may be easily accomplished. Cables and that portion of the connectors which are to be attached to the cables shall not be supplied as part of the equipment. Type 66442 connectors shall be used.

3.3.4 Cases, bases, and mountings. The case sizes and mount loading shall be in accordance with MIL-C-172.

3.3.5 Interchangeability. The equipment shall meet the interchangeability requirements as defined in MIL-E-5400.

3.3.6 Interference control. The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of MIL-STD-461.

3.3.7 Maintenance and field testing. Provisions for maintenance shall be as specified in MIL-E-5400. Specific test points, built-in testing capability as specified in 3.3.15 and test facilities shall be provided for ease of field testing and maintenance. The test points shall be used to isolate a failure to the module level and shall be wired to the external connectors.

3.3.8 Nomenclature and nameplates. Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

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3.3.9 Standard conditions. The following standard conditions shall be used to establish normal performance characteristics and for making laboratory bench tests.

a.	Temperature	Room Ambient (25° $^{+15^{\circ}}_{-5^{\circ}}$ C)
b.	Altitude	Normal ground
c.	Vibration	None
d.	Humidity	Room ambient up to 90% relative humidity
e.	Input Power Voltage	115 ± 2 vac, 400 Hz ± 5 Hz and/or 27.5 ± 1.0 vdc

3.3.10 Service conditions. The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for Class 2 equipment, except as modified herein.

3.3.10.1 Vibration. The equipment shall operate satisfactorily when subjected to the vibration requirements of Curves I and II of MIL-E-5400.

3.3.10.2 Fungus resistance. The requirements of MIL-E-5400 shall apply except that overall spraying of the equipment in accordance with MIL-T-152 shall be made in lieu of fungus treatment of the parts and materials prior to their use in the equipment.

3.3.10.3 Cooling. The equipment shall meet the operational requirements of this specification without external cooling.

3.3.11 Warm-up time. The time required for the equipment to warm-up prior to operation shall be kept to a minimum and shall not be greater than one minute under standard conditions and two minutes at extreme service conditions.

3.3.12 Primary input power requirements. The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power source with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not be greater than the specified amounts.

a.	AC Power (Single Phase)	115-Volt, 400Hz
	Operating Voltage Limits	102 to 124 volts
	Operating Frequency Limits	380 to 420 Hz
	Power (1)	70 V.A. max. (exclusive of power supplied to other equipments)
	Power Factor	0.85 or better

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b.	DC Power	28 VDC
	Operating Voltage Limits	21 to 29 volts
	Power	15 Watts max.

3.3.12.1 Under voltage protection. The equipment shall not be damaged by voltages below the minimum specified herein and shall automatically resume operation when the voltage returns within limits.

3.3.12.2 Overload protection. Overload protection for the equipment shall be provided in the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall have suitable protective devices or shall be proportioned to withstand such overload without permanent damage to the equipment. The use of fuses and other protective devices should be kept to a minimum.

3.3.13 Fittings. The following fittings shall be provided:

a. Pitot fitting. The pitot fitting shall be in accordance with Standard Drawing MS33656-4.

b. Static fitting. The static fitting shall be in accordance with Standard Drawing No, 7603729.

3.3.14 Orientation. Components of the system requiring critical orientation in the aircraft shall be oriented to minimize system errors and meet the requirements of this specification.

3.3.15 Built-in test requirements. The equipment shall contain provisions for built-in testing. Relays, feedback potentiometers, and associated solid state circuitry shall be provided in the impact pressure and altitude modules to simulate sensor signals into the servo amplifiers. The relays shall be activated by an external solid state relay driver. The relays shall be energized by an external +28 vdc, two wire, 75 ma maximum, isolated signal (see Figure 4). Fixed voltages shall be applied to the inputs of the servo amplifiers to simulate an altitude of 20,000 ±100 feet and a Mach number of .75 ± .006. No extra provisions shall be provided for reading out information during built-in testing. The existing potentiometer and synchro outputs shall be used for this purpose. (When the built-in testing is performed, power shall be removed from the fixed windings of the servo motor in the AAU-19/A Servoed Altimeter (see 3.5). This shall be accomplished by applying the external 28 vdc signal to pins 7 and 10 of the AAU-19/A Servoed Altimeter connector as shown in figure 4.)

3.4 Performance. Unless otherwise specified herein, values set forth to establish the requirements for satisfactory performance apply to reduced performance under the extreme conditions is acceptable. Tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified herein.

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3.4.1 Operation. The equipment shall operate to provide outputs of pressure altitude (feet), static pressure (inches Hg), and Mach number derived from the aircraft pitot-static system inputs of indicated static pressure and total pressure (P_{si} and P_t). The following equations define this operation:

a. Subsonic Mach $\quad M = 5 \left[\frac{q_{ci}}{P_{si}} + 1 \right]^{2/7} - 1 \Bigg]^{1/2}$ for $M_i \leq 1.0$

b. Supersonic Mach $\quad \frac{q_{ci}}{P_{si}} = \frac{166.922\ M_i^{\ 7}}{(7M_i^{\ 2} - 1)\ 5/2} - 1$ for $M_i > 1.0$

c. $P_t = P_{ti}$

d. $P_{si} = P_s + \Delta P_s$

e. hp = Pressure Altitude as defined by AFCRC-TR-59267, ARDC Model Atmosphere, 1959 Air Force Surveys in Geophysics No. 115.

f. P_{si} = Indicated Static Pressure

g. P_t = Total Pressure

h. P_s = Static Pressure

i. M_i = Indicated Mach Number

j. q_{ci} = Difference between total pressure and indicated static pressure (indicated impact pressure)

k. q_c = Difference between total pressure and static pressure (impact pressure)

l. $q_{ci} = q_c + \Delta P_s$

m. $q_{ci} = P_{ti} - P_{si}$

3.4.2 Range. The equipment shall be of specified performance throughout the aerodynamic range of the aircraft as shown in Figure 1. Individual transducer ranges shall be equal to or greater than those listed below:

a.	Static Pressure	31.242 IN Hg to 3.264 IN Hg
b.	Altitude	-1200 feet to +51,000 feet
c.	Impact Pressure	0.33 IN Hg to 25.46 IN Hg
d.	Mach Number	0.13 Mach to 1.3 Mach

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3.4.3 Sensitivity. The equipment shall have sufficient sensitivity to satisfy the output requirements throughout the aerodynamic range of the aircraft. Individual transducer sensitivitives shall be equal to or better than those listed below:

a. Static Pressure	±0.002 IN Hg
b. Altitude	±1.5 feet or better at Sea Level ±15 feet or better at 51,000 feet
c. Impact Pressure	±0.002 IN Hg
d. Mach	±0.001 Mach

3.4.4 Accuracy. The equipment shall have sufficient accuracy to satisfy the output requirements throughout the aerodynamic range of the airplane during all flight altitudes.

3.4.4.1 Individual transducer accuracies. Individual transducer accuracies shall be equal to or better than those listed below:

a. Static Pressure	±0.015 IN Hg
b. Impact Pressure	±0.015 IN Hg

3.4.4.2 Computer output accuracies. Computer output accuracies shall be equal to or better than those listed as follows:

a. Altitude	±25 feet or ±0.25% of altitude, which is greater, plus an additional allowance equal to the (±) product of the allowable Mach error and the slope of the static pressure defect curve expressed in feet as shown in figure 2. The additional allowance shall not exceed ±50 feet. The allowable Mach error shall not be greater than ±0.010.
b. Mach Number	See figure 1.

3.4.5 Response. The equipment shall be capable of the following follow-up rates:

a. Static Pressure	0.85 IN Hg/Second minimum as referred to the input.
b. Impact Pressure	0.71 IN Hg/Second minimum as referred to the input ($P_{ti} - P_{si}$).

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c.	Mach Number	0.1 Mach/Second minimum as referred to the input (log q_{ci}/P_{si}).

3.4.6 Total ac power. The total ac primary input power of this equipment shall not be greater than the requirements of 3.3.12(1).

3.4.7 Total dc power. The total dc input power of this equipment shall be as specified in 3.3.12(2). Its primary function will be to energize relays and clutches.

3.4.8 Inputs. The equipment shall be capable of accepting the following inputs which will be provided by associated equipment to enable the air data computer to produce the outputs required (see 3.4.9).

3.4.8.1 Pitot-Static System. Separate inputs of pitot pressure (total pressure) (P_t) and static pressure (P_{si}) will be provided from the pitot-static system of the aircraft.

3.4.9 Outputs. The following outputs shall be provided from the air data computer to other aircraft equipment. The detail descriptions and functions of the various outputs are described as follows:

a. Output 1, 2, 3, 4, 5, and 6 - Impact Pressure (Autopilot).

(1)	Type	Linear Rheostat
(2)	Resistance	50,000 ohms ±5% (from 1.414 IN Hg to 25.46 IN Hg)
(3)	Power Rating	1 Watt at +71°C
(4)	Absolute Conformity	0.35%
(5)	Resolution	0.1% minimum
(6)	Range	0.33 IN Hg to 25.46 IN Hg
(7)	Scale Factor (Rheostat)	Above 1.414 IN Hg 2079.347916 ohms/IN Hg applies

b. Output Conformity. The output of the impact pressure rheostat shall be less than 100 ohms for all values of q_c less than 1.414 IN Hg.

c. Outputs 8 and 9 Static Pressure (Power Trim Computer).

(1)	Type	2 Linear Rheostats
(2)	Current Rating	50 Milliamps
(3)	Absolute Resistance Conformity	±5.00 ohms

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(4)	Resolution	±0.25%
(5)	Range	18.324 IN Hg to 31.242 IN Hg
(6)	Scale Factor	34.3802 ohms/IN Hg

d. Output Conformity. The output of each rheostat shall be linear with static pressure changes. The resistance between the terminals shall be less than 1 ohm for all values of P_{si} below 18.342 IN Hg.

e. Output 10 Static Pressure (A/D Converter).

(1)	Type	Control Transmitter, MIL-S-20708/8B (26V - 11CX4c)
(2)	Range	3.264 to 31.242 IN Hg 0 degrees at 0 IN Hg
(3)	Scale Factor	11.25 degrees/IN Hg Electrical zero at 0 IN Hg (see Figure 3)
(4)	Excitation Voltage	26 vac, 400 Hz (supplied by A-D Converter)
(5)	Phase Shift	0 5 degrees
(6)	Null Voltage	19 MV max. total 12 MV max. fundamental
(7)	Electrical Error	7 minutes max.

f. Output 12 Altitude (AIMSO).

(1)	Type	Synchro Control Transformer Bendix Type No. AY-503-3-A1 or equiv.
(2)	Scale Factor	36 per 1,000 feet Electrical Zero at 0.0 feet (see Figure 3)
(3)	Range	-1,200 feet to 51,000 feet
(4)	Phase Shift	0 10 degrees
(5)	Null Voltage	35 MV max. total 22 MV max. fundamental
(6)	Electrical Error	15 minutes max.

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g. Output 13 Altitude (AIMS).

(1) Type	Analog to Digital Altitude Reporting Encoder (Datex Type 11-105-3 or equivalent)
(2) Scale Factor	100 feet per count, each increment to be nominally centered at even 100 foot increments of pressure altitude. The specified 100 foot increments dictate the switching points being at 50 feet removed from the indicating altitude. Accordingly the points at which the encoder is checked will be offset by 50 feet on the test equipment.
(3) Range	-1,200 feet to 51,000 feet
(4) Lead Wire Colors	As in Table I

h. Output 14 Altitude Control (Barometric Altitude Hold [Autopilot]).

(1) Type	Two wire isolated 400 Hz clutched synchro output proportional to the deviation from the "engage" altitude with phase reversing to indicate directions of departure.
(2) Range	±480 feet
(3) Scale Factor	12.5 MV/ft. to 45,000 ft., above 45,000 ft., linear change from 12.5 MV/ft. to 6.0 MV/ft. at 51,000 feet.
(4) Sensitivity	±1.5 feet or better at sea level ±15 ft. or better at 51,000 feet.
(5) Excitation Voltage	26V, 400 Hz (suppled by autopilot)
(6) Phase Shift	0° ±10°
(7) Linearity	The linearity deviation of the synchro within ±60° of null shall not be greater than ±1%.

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TABLE I. Identification of lead wires

Encoder Lead	Color
Common	Black
D2	Brown
D4	Red
A1	Orange
A2	Yellow
A4	Green
B1	Blue
B2	Violet
B4	Grey
C1	White
C2	White-Red Tracer
C4	White-Brown Tracer
Case Ground if necessary	White-Black Tracer

i. Output 15 Altitude/Mach Module Monitor.

(1) Type — Switch, Single Pole, Double Throw

(2) Activating Signal — (a) Loss of power to computer
(b) Excessive servo error signal corresponding to 0.1 IN Hg
(c) Excessive servo error signal corresponding to 5 to 15% of indicated Mach number.

(3) Time Delay — 3 seconds max.

j. Outputs 17 and 18 Mach Number (Autopilot).

(1) Type — Two linear potentiometers.

(2) Resistance — 5000 ohms ±5%

(3) Power — 1 Watt at +71°C

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(4)	Absolute Conformity	0.35%
(5)	Resolution	0.1% or better
(6)	Scale Factor	0.1 VR at .13 Mach; 0 VR at 0 Mach; 1 VR at 1.3 Mach

k. Output 19 Mach Number (A/D Converter).

(1)	Type	Control Transmitter MIL-S-20708/8B (26V - 11CX4c)
(2)	Range	0.13 to 1.3 Mach 0 degrees at 0 Mach
(3)	Scale Factor	257.1 degrees/Mach Electrical zero at 0 Mach (see Figure 3)
(4)	Excitation Voltage	26 vac, 400 Hz (supplied by A/D Converter)
(5)	Phase Shift	0 ±5 degrees
(6)	Null Voltage	19 MV max Total, 12 MV max Fundamental
(7)	Electrical Error	0 ±7 minutes max

l. Output 20 Mach Control [(Incremental Mach) Autopilot].

(1)	Type	Type wire isolated 400 Hz clutched synchro output proportional to the deviation from the "engage" Mach with phase reversing to indicate direction of departure.
(2)	Range	±0.05 Mach
(3)	Scale Factor	0.079 Volts/0.001 Mach
(4)	Sensitivity	0.001 Mach
(5)	Excitation Voltage	26V, 400 Hz (supplied by autopilot)
(6)	Phase Shift	0° ± 10°
(7)	Linearity	The linearity deviation of the synchro within ±60° of null shall not be greater than ±1%.

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(8)	Engage Clutch	The clutch shall be engaged by the application of 28 + 2 volts dc - 7 from a remote source for Mach hold operation. When the voltage is removed, a mechanism shall return the output device to its null position.
(9)	Null	0.045 volts total or less before and after clutch engagement.

m. Output 21 Fault Isolation (Course Logqc).

(1)	Type	Synchro, control transmitter, Bendix Type No. AY 503-25-A7 or Equivalent.
(2)	Scale Factor	$240° - \left[75° \text{ Log}\left(\frac{q_c}{0.27012}\right)\right]$
(3)	Range	0.33 IN Hg to 25.46 IN Hg

n. Output 22 Fault Isolation (Fine Log).

(1)	Type	Synchro, control transmitter Bendix Type No. AY-500-25-A7 or equivalent
(2)	Scale Factor	$240° - \left[675° \text{ Log}\left(\frac{q_c}{0.27012}\right)\right]$
(3)	Range	

o. Output 23 Fault Isolation (Coarse Log /).

(1)	Type	Synchro, differential transmitter Bendix Type No. AY-533-25-A2 or equivalent
(2)	Scale Factor	$60° - \left[75° \text{ Log}\left(\frac{q_c}{0.27012}\right)\right] - 75°\left[\text{Log}\left(\frac{P_s}{2.11778}\right)\right]$
(3)	Range	31.242 IN Hg to 3.264 IN Hg

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p. Output 24 Fault Isolation (fine Log qc/Ps).

(1) Type	Synchro, differential transmitter Bendix Type No. AY-533-25-A2 or equivalent
(2) Scale Factor	$60° - \left[675° \left(\text{Log} \left(\frac{q_c}{0.27012} \right) \right) \right] - 675° \left[\text{Log} \left(\frac{P_s}{2.11778} \right) \right]$
(3) Range	31.242 IN Hg to 3.264 IN Hg

q. Output 25 AAU/19A Power.

(1) Voltage	102 to 124 volts
(2) Frequency	380 to 420 Hz
(3) Power (1Ø)	30 V.A. max

3.5 Detail requirements.

3.5.1 Rack assembly. The rack assembly shall be in accordance with standard Air Transport Rack (ATR) provisions except for the envelope size, and except that the modules (see 3.5.2) shall be mounted on a flat tray and shall meet the following requirements:

3.5.1.1 Function. The rack assembly shall provide for the following:

a. Plug-in mounting of the module subassemblies on one tray.

b. Interconnecting wiring between the module connectors on the tray and the external connectors on the front panel.

c. Mounting of the air data computer in the aircraft and isolation of the equipment from external vibration.

d. Mounting of the elapsed time indicator, static and pitot fittings, and electrical connectors on the front panel.

3.5.1.2 Form factor. The form factor and outline dimensions shall not be greater than those shown in Drawing No. 560T58-1.

3.5.1.3 Contents. The rack assembly shall consist of the following:

a. Mounting Rail Assembly

b. Shock Mount Assembly

c. Module Mounting Assembly

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d. Electrical connections

e. Static and pitot ports

f. Elasped Time Indicator

g. Relay for Built-In-Test

3.5.1.3.1 Mounting rail assembly. The mounting rail shall be designed for fixed mounting in the aircraft and to provide mounting for the shock mounting assembly.

3.5.1.3.2 Shock mount assembly. The shock mount assembly shall mount on the mounting rail and shall provide the following:

a. Mounting rack

b. Shock mounts (4)

c. Mounting provisions for the module mounting assembly

d. Interconnection tubing between external pressure ports and the modules

e. Interconnecting wiring between the external electrical connectors and the modules

3.5.1.3.3 Module mounting assembly. The module mounting assembly shall mount on the shock mount assembly and shall provide the following:

a. Provisions for securing the plug-in module.

b. Provisions for mounting the internal (module) connectors of the shock mount assembly.

c. Provide quick disconnect pressure connections for the altitude and impact pressure modules.

3.5.1.3.4 Electrical connections. The rack assembly shall provide external connectors.

3.5.1.3.5 Elapsed time indicators. The rack assembly shall provide a 10,000 hour numerical readout elapsed time indicator in accordance with MIL-M-7793. The time indication shall be visible through the computer housing.

3.5.2 Modules. The following modules shall be provided as plug-in assemblies to supply the electro-mechanical devices required to meet the requirements of 3.3. The form factor and sway limitations for the rack assembly and modules shall be in accordance with Drawing No. 560T58-1.

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3.5.2.1 General.

3.5.2.1.1 Cases. The case of each module shall be environmentally sealed and finished in lusterless black, Color 37038, in accordance with TT-L-20. The seal shall be accomplished in a manner that will not be dependent upon materials that will be affected by the action of any atmosphere to which the case will be subjected in the tests specified herein.

3.5.2.1.2 Mounting. Provisions for mounting of the modules shall be compatible with the mounting provisions on the module mounting assembly.

3.5.2.1.3 Electrical connections. Each module shall provide electrical connections to a connector on the bottom of the case for plugging into the tray of the module mounting assembly.

3.5.2.1.4 Interchangeability. All similar modules shall be interchangeable on any air data computer without affecting performance of the air data computer.

3.5.2.1.5 Pressure connections. The impact pressure and altitude modules shall each have a quick disconnect pressure port in its base.

3.5.2.2 Impact pressure computer module. The impact pressure module shall meet the following requirements.

3.5.2.2.1 Function. The impact pressure module shall provide for computation of impact pressure and all other functions as required by the various air data computer modules.

3.5.2.2.2 Form factor. The form factor of this module added to form factors of the other two (2) modules shall not be greater than maximum dimensions shown in Drawing No. 560T58-1.

3.5.2.2.3 Contents. The impact pressure module shall consist of:

a. Case (see 3.5.2.1.1)

b. Pressure Sensor

c. Impact Pressure Servo Amplifier

d. Additional Circuitry as required

e. Electrical Connections

f. Provisions for Static Defect Compensation

g. Function Generators

3.5.2.2.3.1 Function generator. The module shall be capable of converting impact pressure to log impact pressure.

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3.5.2.2.3.2 Static defect compensation. Impact pressure error shall be provided from a potentiometer in the altitude module. This error shall be summed with the transducer output to provide a corrected output at the impact pressure shaft.

3.5.2.2.3.3 Pressure sensor. The sensor shall be of the type which contains no mechanical pivots or linkages or flexures in transferring capsule information to the electrical pickup. This provision is essential in order to minimize friction and assure high sensitivity as specified herein.

a. Pitot volume. The total pitot volume of the computer shall not be greater than 5 cubic inches.

b. Static volume. The static volume of the pressure altitude and impact pressure modules shall not be greater than 5 cubic inches each.

3.5.2.2.3.4 Impact pressure servo amplifier. A transistorized servo amplifier shall be provided in the impact pressure module to amplify signals from the differential pressure transducer to produce useable servo motor voltage.

3.5.2.2.3.5 Additional circuitry. The module shall provide all other associated impact pressure computer circuitry.

3.5.2.3 Mach number module. The Mach Number module shall meet the following requirements:

3.5.2.3.1 Function. The Mach Number module shall compute Mach using the output (Log q_c minus Log P_s) from the pressure altitude module synchro differentials.

3.5.2.3.2 Form factor. The form factor of this module added to the form factors of the other two modules shall not be greater than the maximum dimensions shown in Drawing No. 560T58-1.

3.5.2.3.3 Contents. The Mach Number module shall consist of:

a. Case (see 3.5.2.1.1)

b. Mach Servo Amplifier

c. Additional circuitry as required

d. Electrical Connections

e. Provisions for Static Defect Compensation

f. Function Generator

g. AIMS Failure Detection Circuitry

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3.5.2.3.3.1 Mach servo amplifier. A transistorized servo amplifier shall be provided to amplify the signals required to produce a useable servo motor voltage.

3.5.2.3.3.2 Additional circuitry. The module shall provide all other associated Mach Number module computer circuitry.

3.5.2.3.3.3 Function generator. The module shall be capable of converting log impact pressure minus log static pressure to Mach number.

3.5.2.3.3.4 Static pressure defect compensation. The unit shall have a functionalized potentiometer driven by the mach shaft. This potentiometer will supply error information in the form of $\Delta \frac{P}{P_s}$. The output from this potentiometer shall be used to excite a potentiometer in the pressure altitude module. Correction will be as per Figure 2.

3.5.2.3.3.5 AIMS failure detection circuitry. The module shall contain a monitor which will activate two switches simultaneously in the event of excessive servo error signal corresponding to 5 to 15 percent of indicated Mach number. One switch shall open the common lead of Output No. 13 and the high side of Output No. 25 (115V 400 Hz power to the AAU-19/A Servoed Altimeter). The other switch shall be a set of single pole double throw contacts isolated from ground, Output No. 15 (Altitude/Mach Monitor). This switch shall be available for use by other equipments which require altitude failure warning. A time delay shall be incorporated into the monitor to prevent nuisance trips from power fluctuation or pressure disturbances as encountered in turbulent air conditions. The time delay shall not be greater than three seconds.

3.5.2.4 Pressure altitude module. The pressure altitude module shall meet the following requirements.

3.5.2.4.1 Function. The pressure altitude module shall compute all static pressure and pressure altitude outputs plus all other functions as required by the various air data computer modules.

3.5.2.4.2 Form factor. The form factor of this module added to the form factor of the other two (2) modules shall not be greater than the maximum dimensions shown in Drawing No. 560T58-1.

3.5.2.4.3 Contents. The pressure altitude module shall consist of the following:

a. Case (see 3.5.2.1.1)

b. Pressure Sensor

c. Altitude Servo Amplifier

d. Additional Circuitry as required

e. Electrical Connections

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f. Provisions for Static Defect Compensation

g. Function Generators

h. AIMS Failure Detection Circuitry

3.5.3.4.3.1 Pressure Sensor. The sensor shall be of the type which contains no mechanical pivots, linkages, or flexures in transferring capsule information to the electrical pickup. This provision is essential in order to minimize friction and assure high sensitivity as specified herein.

3.5.2.4.3.2 Altitude Servo Amplifier. A transistorized servo amplifier shall be provided in the pressure altitude computer module to amplify signals from the static pressure transducer to produce useable servo motor voltage.

3.5.2.4.3.3 Additional Circuitry. The module shall provide all other associated pressure altitude computer circuitry.

3.5.2.4.3.4 Function Generators. The module shall be capable of converting static pressure information to altitude and log static pressure.

3.5.2.4.3.5 Static Defect Compensation. The unit shall have a potentiometer driven by the pressure shaft. This potentiometer shall convert the error to $\frac{\Delta P_s}{P_s}$ The static pressure error shall be electrically summed with the transducer output to provide corrected outputs. Impact pressure error shall be supplied to the impact pressure module. Correction will be as per figure 2.

3.5.2.4.3.6 AIMS Failure Detection Circuitry. The module shall contain a monitor which will activate two switches (the same switches referred to in 3.5.2.3.3.5) simultaneously in the event of failure of 115V 400 Hz power to the air data computer, or in the event of excessive servo error signal (equal to a pressure altitude of 0.1 inches of Hg). One switch shall open the common lead of Output No. 13 and the high side of the Output No. 25 (115V 400 Hz power to the AAU-19/A Servoed Altimeter). The other switch shall be a set of single pole double throw contacts, isolated from ground, Output No. 15. This switch shall be available for use by other equipments which require altitude failure warning. A time delay shall be incorporated into the monitor to prevent nuisance trips from power fluctuations or pressure disturbances as encountered in turbulent air conditions. The time delay shall not be greater than three seconds.

3.6 Precedence of Documents. When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

a. Contract. The contract shall have precedence over any specification.

b. This Specification. This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification or subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

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c. Referenced Specifications. Any referenced specification shall have precedence over all applicable subsidiary specifications reference therein. All referenced specifications shall apply to the extend specified.

d. Reordered Equipment. The following specifications shall take precedence over those concerning performance requirements listed in 2.1, where equipment is reordered.

MIL-E-5400B	27 May 1957
MIL-T-5422E	15 Nov 1961
MIL-STD-461	1 Aug 1968

3.7 Workmanship. The test set shall conform in every respect to the requirements of this specification and applicable paragraphs of MIL-E-5400.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specificed herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure supplies and service conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Preproduction (First article) Inspections (see 4.4)

b. Initial production Inspections (see 4.5)

c. Quality Conformance (Acceptance) Inspections (see 4.6)

d. Life test (see 4.8)

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the conditions specified in 3.3.9 and 3.3.10.

4.3.1 Performance characterics.

4.3.1.1 Before testing. The equipment shall satisfy the requirements for interference control (see 3.3.6) and warm-up time (see 3.3.11).

4.3.1.2 During inspection acceptance testing. The equipment shall satisfy the requirements of Section 3.

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First article test performance shall be the responsibility of the contractor and shall be performed in accordance with approved test procedures (see 4.6). The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all the required tests shall obtain the services of a commercial laboratory acceptable to the Government. No preproduction inspections shall be conducted prior to approval of the preproduction test procedure by the procuring activity (see 6.2).

4.4.2 First article test data. The contractor shall prepare all data, (see 6.2), collected in conducting these tests for procuring agency review and approval. On first production contracts, this data shall include a list of all electrical and electronic parts giving their specified voltage, and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

4.4.3 Scope of first article tests. First article tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and contract.

4.4.4 First article approval. Approval of the first article model shall be by the procuring activity upon satisfactory completion of all tests and first article test report made available (see 6.2.2). No production equipments shall be delivered prior to the approval of the first article model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved first article model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The first article model shall not be considered as one of the equipments under the contract.

4.4.5 Production equipments. Equipments supplied under the contract shall in all respect, including design, construction, workmanship, performance, and quality, be equal to the approved first article sample (see 6.2.2). Each equipment shall be capable of successfully passing the same tests imposed on the first article sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government. It shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.5 Initial production sample tests. When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests (see 6.2.1). The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests.

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4.5.1 Scope of tests. This equipment shall be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equal to the previously approved preproduction sample in design, construction, workmanship, performance, and quality and that it meets all applicable requirements.

4.5.2 Accessory material. In addition to complete equipment for initial production tests, the contractor shall also have available the accessory material and design and test data specified in the contract (see 6.2.2). This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.5.3 Initial production sample approval. Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests and preparation of an initial production test report. Any design, material or performance, defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity (see 6.2.2).

4.5.4 Reconditioning of initial production test sample. On completion of the initial production sample tests, the equipment shall be reworked by the contractor by replacing all worn or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.6 Acceptance tests. The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests in accordance with Document No. 7102432, Part I. All inspection and testing shall be to the approved acceptance test procedures under the supervision of the Government inspector (see 6.2.2). Contractors not having testing facilities acceptable to the procuring activity shall engage the services of a commercial testing laboratory approved by the procuring activity. The contractor's test reports shall show quantitative results for all acceptance tests (see 6.2.2). Such reports shall be signed by an authorized representative of the contractor and laboratory as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of acceptance of the finished product. Acceptance test shall consist of the following:

a. Individual tests (see 4.6.1)

b. Sampling tests (see 4.6.2) or Reliability assurance tests (see 4.6.3)

c. Special tests (see 4.6.4)

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4.6.1 Individual tests. Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy, and reliability. As a minimum, each equipment accepted shall have passed the following tests:

a. Examination of product

b. Operational test

c. Manufacturing burn-in tests

4.6.1.1 Examination of product. Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.6.1.2 Operational test. Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.6.1.3 Manufacturing burn-in test. Each equipment shall be operated under the conditions specified in the approved Burn-In Procedure, Document No. 7102530 for a minimum period of 72 hours. A failure shall be defined as anything which causes malfunctioning of the equipment.

4.6.2 Sampling tests. Sampling tests shall be conducted on equipments which have successfully passed the individual and burn-in tests. Equipments shall be selected for sampling tests by the Government inspector in accordance with Table II.

TABLE II. Sampling test selection.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 100	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.6.2.1 Scope of tests. As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

a. Complete operational tests at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

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b. Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, production, and special tests.

c. Manufacturing burn-in test specified in 4.6.1.3.

4.6.3 Reliability assurance tests. Reliability Assurance Tests shall be conducted in accordance with MIL-STD-785. Tests as required by both the Qualification Phase and the Production Acceptance (Sampling) Phase shall be conducted. Classification of failures shall be in accordance with MIL-STD-781 and MIL-STD-2074.

4.6.3.1 Qualification phase. Prior to the acceptance of equipments under the contract or purchase order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level "F" shall be used. The Accept-Reject Criteria for Test Plan IV shall be used.

4.6.3.2 Reliability production acceptance (Sampling) phase tests. The equipment, throughout production, shall be tested as outlined in MIL-STD-781 (as modified herein) under the section entitled "Production Acceptance (Sampling) Phase of Production Reliability Tests". Test level "F" shall be used.

4.6.3.2.1 All equipment test. Each equipment produced, except those submitted for the Reliability Qualification Test, shall be tested for 48 operating hours. Prior to the 48 hour test on each equipment, if a burn-in period is to be used, the details thereof must be included in the approved test procedures. To determine whether the MTBF is being met at any time during the contract, the operating test hours and the failures thereon (not counting burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of Test Plan II of MIL-STD-781. (Extend the line as necessary to accommodate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month, test results shall be available to the procuring activity (see 6.2.2). At any time that the current totals of test hours and test failures plotted on Test Plan II curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision.

4.6.3.3 Test details. The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be available for approval by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests (see 6.2.2).

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4.6.4 Special tests. Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests. (When a Production Sample is requested and delivered to a government laboratory for test (see 4.3) the requirement for a special test on an early production equipment may be omitted.)

4.6.4.1 Special tests schedule. Selection of equipment for special tests shall be made as follows:

a. On an early production equipment.

b. On an early equipment after a design or material change.

c. Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity.)

4.6.4.2 Scope of tests. Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure for the procuring activity approval prior to conducting the tests (see 6.2.2). As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

a. Temperature - altitude tests

b. Vibration test in accordance with MIL-E-5400 with the following exception (60 minutes in each of three axes)

c. Humidity tests

4.7 Equipment failure. Should a failure occur during either the sampling or special tests, the following action shall be taken:

a. Determine the cause of the failure.

b. Determine if the failure is an isolated case or design defect.

c. Prepare for the procuring activity approval proposed corrective action intended to reduce the possibility of the same failure(s) recurring in future tests (see 6.2.2).

d. Include a test in the acceptance test to check all equipment until reasonable assurance is obtained that the defect has been corrected.

4.8 Life test. The contractor shall furnish all supplies and shall be responsible for accomplishing the life test. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual

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test. The life test shall be performed under the conditions specified in 4.8.1. The life test sample shall be selected by the Government inspector in accordance with table III, except that when a production sample equipment is requested and submitted to the procuring activity for test then the first equipment specified in table III need not be selected or subjected to this test. (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

TABLE III. Life test sample selection.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25 Next 175 Next 300	1 1 1 1 for each additional 500 or fraction thereof

4.8.1 Test conditions. The life test shall be conducted under the following simulated service conditions.

a.	Temperature	Normal Room
b.	Altitude	Normal Ground (0-5000)
c.	Humidity	Room Ambient
d.	AC Voltage	115 ±5 volts (at lowest applicable frequency)
e.	DC Voltage	27 ±2 volts

4.8.2 Test periods. The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation. In addition, the pitot and static inputs of the equipment shall be pneumatically cycled during the entire life test. A cycle shall consist of operation from minimum to maximum altitude and mach number inputs and return to minimum within a period of approximately 15 minutes. A minimum of 3 cycles per hour of operation shall be performed.

4.8.3 Performance check. At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity (see 6.2.2).

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4.8.4 Test data. The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test (see 6.2.2). This record shall indicate the following:

a. Part Number

b. The circuit reference symbol number

c. The part function

d. Name of the manufacturer

e. Nature of the failure

f. The number of hours which the part operated prior to failure.

4.8.4.1 Failure report. In the event of a failure, the Government inspector shall be notified immediately. A report shall be prepared for the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable (see 6.2.2).

4.9 Reconditioning of test samples. An equipment which has been subjected to acceptance or life tests shall be reconditioned as follows:

a. Upon completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

b. After reworking, the contractor may resubmit the equipment for acceptance.

4.10 Presubmission testing. No item, part, or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply with all applicable requirements.

4.11 Rejection and retest. Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be made available to the Government inspector (see 6.2.2).

4.12 Test Procedures. The procedures used for conducting preproduction tests, acceptance tests, and life tests shall be prepared by the contractor for the procuring activity review and approval (see 6.2.2). The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract (see 6.2.1).

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MIL-T-18303 shall be used for preparation of test procedures. When the procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

5. PACKAGING.

5.1 General. Packaging shall be in accordance with the general requirements of MIL-E-17555, supplemented as follows:

5.2 Preservation-packaging. Unless otherwise specified in the contract or purchase order, preservation-packaging shall be in accordance with MIL-P-116, Method III (see 6.2.1).

5.3 Packing. Unless otherwise specified in the contract, or purchase order, packing shall be level A. The packaged equipments shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, normal requirements. Sealing (taping) and reinforcing (banding) shall be in accordance with the appendix to PPP-B-636 (see 6.2.1).

5.3.1 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract, shall be used whenever the total quantities for shipment to one destination equal 40 cubic feet or more. Unitized loads shall be uniform in size and quantities to the greatest extent possible.

5.4 Marking. In addition to any special marking required by the contract (see 6.2); each unit package, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

6. NOTES.

6.1 Intended use. The air data computer will be used to compute data required by associated systems in the aircraft.

6.2 Ordering data.

6.2.1 Procurement requirements. Purchasers should exercise any desired options offered herein; and procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable levels of packaging and packing (see Section 5).

c. When a first article is required for inspection and approval (see 4.4).

d. On reorder, indicate if previously approved reliability program may be continued (see 3.3.2).

e. When initial production tests are required (see 4.5).

f. If modification of test procedures is required (see 4.12).

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6.2.2 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9 (n) of the Defense Acquisition Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
3.3.2.2	Report, Repair Parts	UDI-L-26529
4.4.1, 4.12	Procedure, First Article Test	DI-T-4901
4.4.4, 4.4.5	Report, First Article Test	DI-T-4902
4.5.2	Procedure, Initial Production Test	DI-T-4903
4.5.3	Report, Initial Production Test	DI-T-4904
4.6, 4.12	Procedure, Acceptance Test	DI-T-4903
4.6	Report, Acceptance Test	DI-T-4904
4.6.3.2.1	Report, Reliability (monthly)	DI-R-7035
4.6.3.3	Procedure, Reliability Assurance Test	DI-R-7035
4.6.4.2	Procedure, Special Test	DI-T-4903
3.3	Engineering Change Proposal	DI-E-2037
4.7	Proposal, Corrective Action	UDI-T-23719
4.8.3, 4.12	Procedure, Life Test	DI-T-4903
4.8.4	Report, Life Test	DI-T-4904
4.8.4.1	Report, Failure	UDI-R-21141A
4.11	Report, Corrective Action	UDI-T-23719

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

6.3 Instructions for first article. When a first article is required for inspection and approval (see 3.1, 4.4, 6.3.1).

6.3.1 First article. One preproduction unit will be tested in accordance with 4.4.

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6.4 Test values. Normal and limiting values of performance data shall be determined at input voltages of 27.5 ±0.5 vdc and 115 ±2.0 vac as applicable. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standards of performance.

6.5 Performance objectives. Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which should be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts, and processes other than those specified in MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval should be submitted to the procured activity for consideration. Each request shall be accompanied by complete supporting information.

6.6 Design objectives. As a general rule, nonrepairable subassemblies should be encapsulated or hermetically sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the nonrepairable subassembly is greater than that of similar repairable one. With few exceptions (such as high voltage power supplies) the nonrepairable subassembly should evidence a mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.7 Associated equipment. The equipment should operate with the following associated equipment which should not be supplied as part of the equipment:

a. A/D Converter

b. Automatic Flight Control System, AN/ASW-16

c. Engine Performance Indicating System

d. AAU-19/A or AAU-19A/A Servoed Altimeter

e. KY-553/A Transponder

f. KIT-1/TSEC Computer

Preparing Activity:
Navy - AS

(Project 6610-N019)

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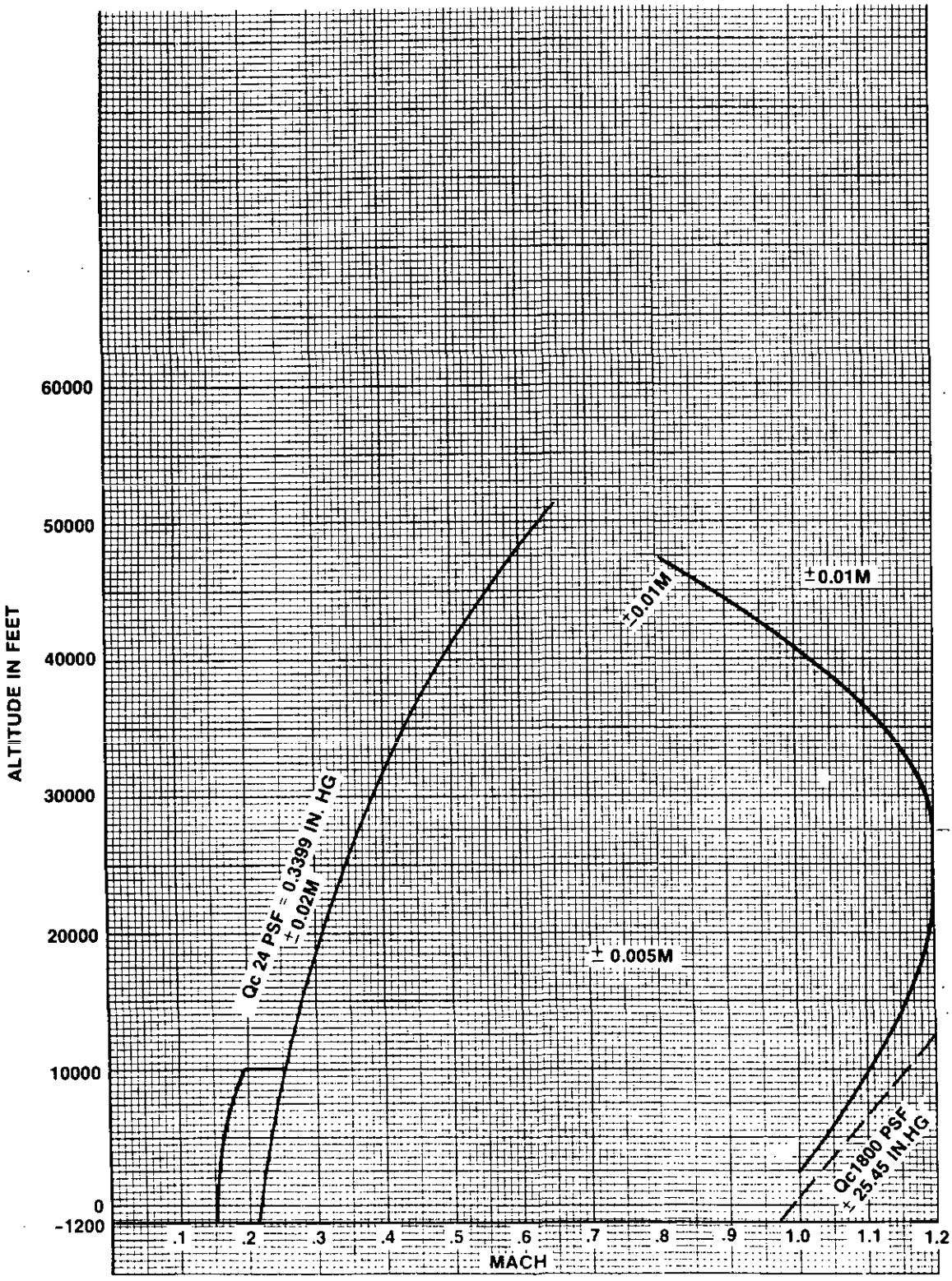
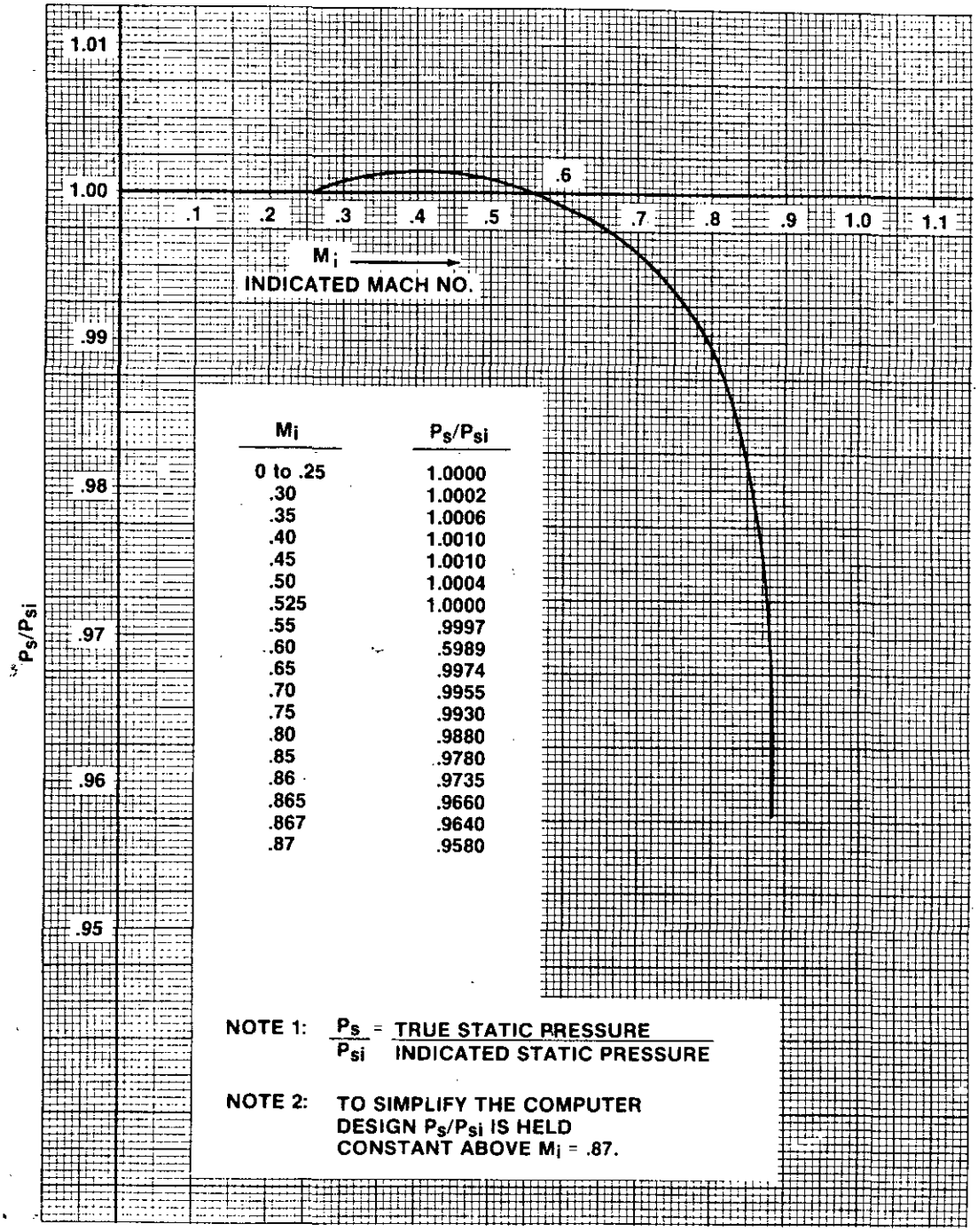


FIGURE 1. MACH accuracies.

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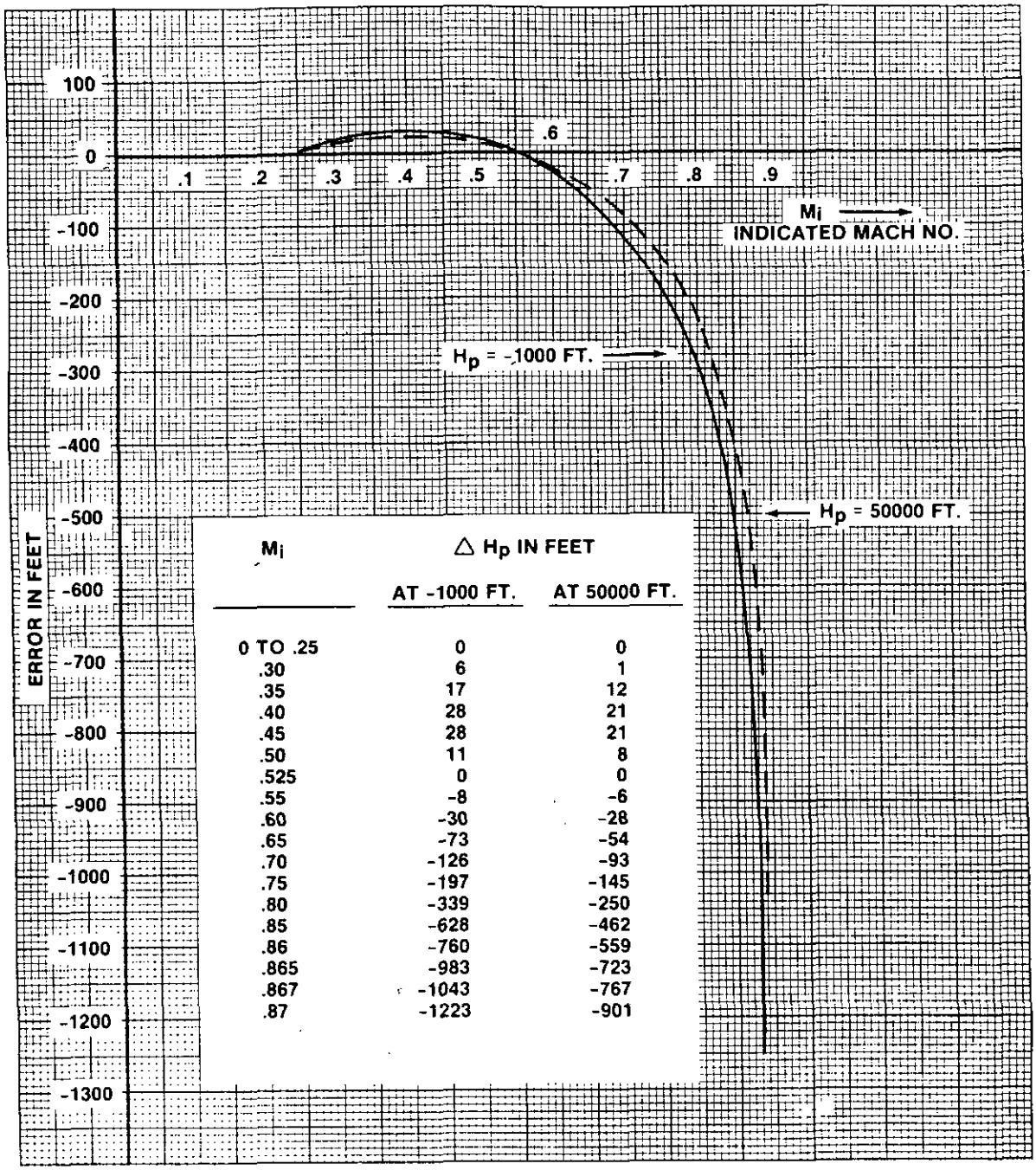
M_i	P_S/P_{si}
0 to .25	1.0000
.30	1.0002
.35	1.0006
.40	1.0010
.45	1.0010
.50	1.0004
.525	1.0000
.55	.9997
.60	.5989
.65	.9974
.70	.9955
.75	.9930
.80	.9880
.85	.9780
.86	.9735
.865	.9660
.867	.9640
.87	.9580

NOTE 1: $\frac{P_S}{P_{si}} = \frac{\text{TRUE STATIC PRESSURE}}{\text{INDICATED STATIC PRESSURE}}$

NOTE 2: TO SIMPLIFY THE COMPUTER DESIGN P_S/P_{si} IS HELD CONSTANT ABOVE M_i = .87.

FIGURE 2. Static pressure defect curve. (sheet 1 of 2)

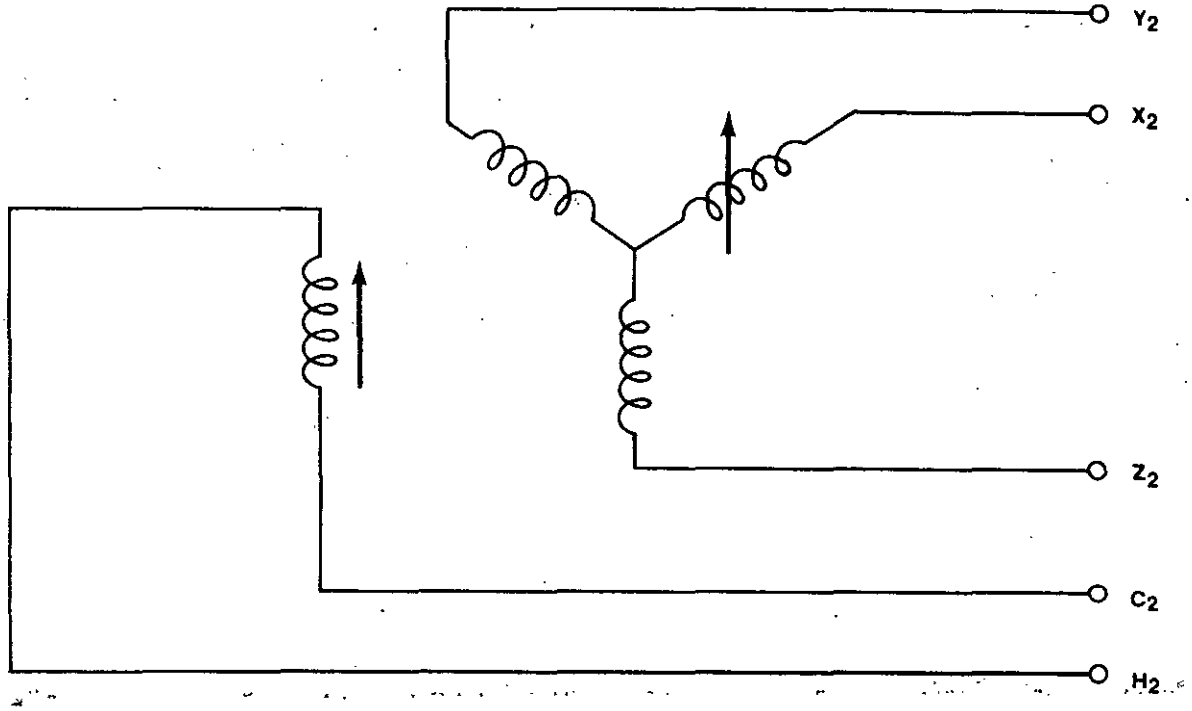
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M_i	△ H_p IN FEET	
	AT -1000 FT.	AT 50000 FT.
0 TO .25	0	0
.30	6	1
.35	17	12
.40	28	21
.45	28	21
.50	11	8
.525	0	0
.55	-8	-6
.60	-30	-28
.65	-73	-54
.70	-126	-93
.75	-197	-145
.80	-339	-250
.85	-628	-462
.86	-760	-559
.865	-983	-723
.867	-1043	-767
.87	-1223	-901

FIGURE 2. Static pressure defect curve. (sheet 2 of 2)

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A. THE ROTOR SHALL BE EXCITED WITH RATED VOLTAGE SO THAT VOLTAGE C_2 TO H_2 IS IN PHASE WITH POWER EXCITATION GROUND TO PHASE.

B. AT THE ELECTRICAL ZERO POSITION, THE VOLTAGE FROM Z_2 TO X_2 AND FROM Z_2 TO Y_2 SHALL BE IN PHASE WITH C_2 TO H_2 AND THE POWER EXCITATION. VOLTAGE X_2 TO Y_2 SHALL BE NULL. IF C_2 IS CONNECTED TO Z_2 THEN THE VOLTAGE H_2 TO X_2 IS LESS THAN H_2 TO C_2.

C. ROTATION OF THE SYNCHRO TO AN INCREASING FUNCTION (SCHEMATICALLY SHOWN AS SYNCHRO ROTOR CLOCKWISE) THE VOLTAGE Y_2 TO X_2 SHALL INCREASE IN PHASE WITH C_2 TO H_2 AND THE POWER EXCITATION.

D. THE X_2, Y_2, Z_2, H_2, AND C_2 DESIGNATIONS WILL BE ASSIGNED TO THE WIRING DIAGRAM OF THE COMPONENT CONNECTOR.

FIGURE 3. Electrical zero transmitter - outputs 10, 12 and 19.

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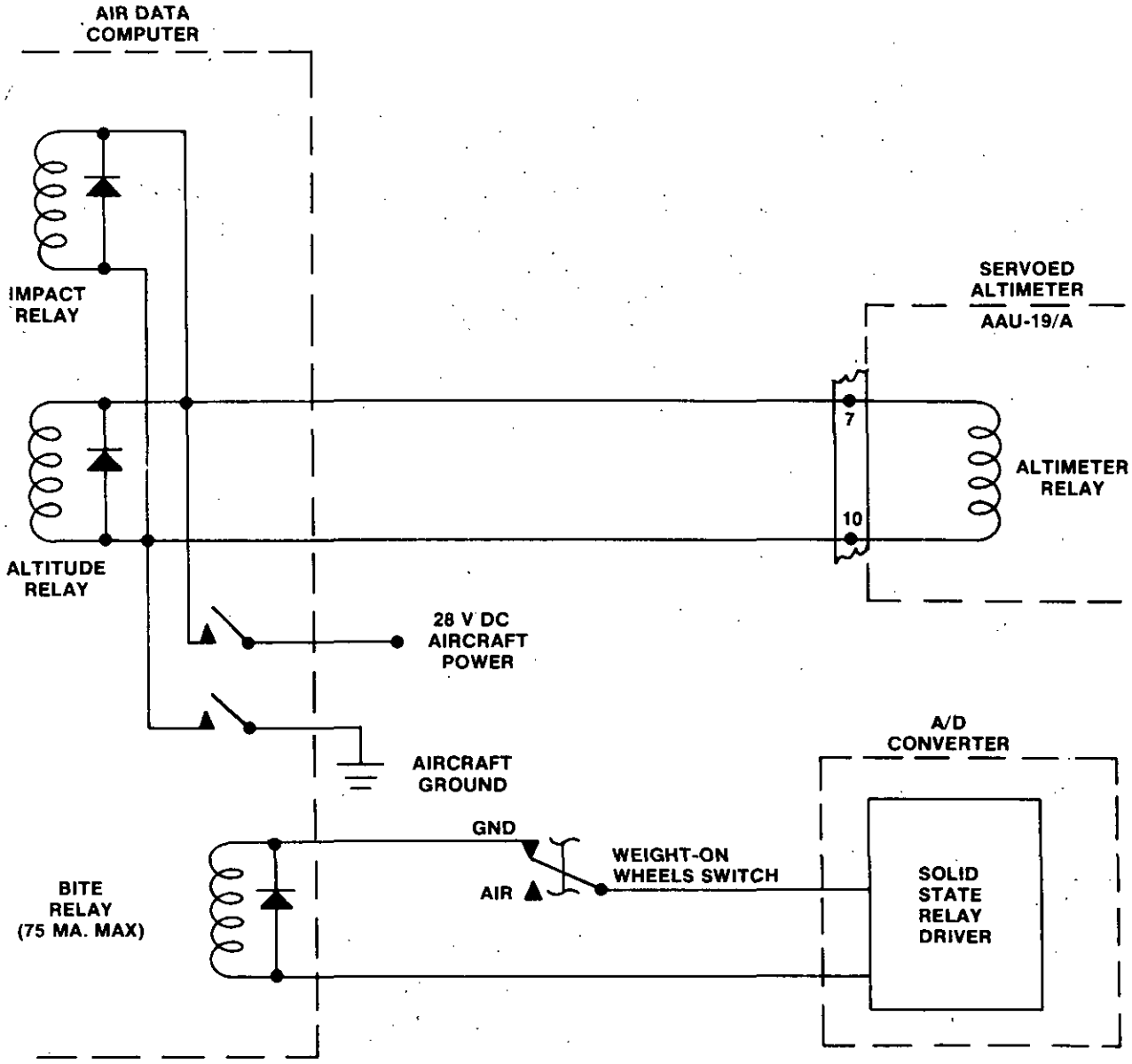


FIGURE 4. Built-in-test interface.

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